Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Driven Digital Twin Reliability Platform Integrating Neutron-Based Non-Destructive Testing for Advanced Semiconductor Devices			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Equipment	*Measurement/Inspection Equipment*		
Concept of Proposed Technology	The proposed technology is an AI-driven digital twin reliability platform that integrates compact neutron-beam non-destructive testing (NDT) with device-level electrical characterization and physics-based modeling. The platform enables systematic analysis of radiation-induced defects, electrical noise, and degradation behaviors in advanced semiconductor devices such as 1D (e.g. SWNT)/2D (e.g.,TMDCs and others) FETs, hafnia-based ferroelectric 2-terminal (memristor type) and 3-terminal devices (FET type), and RRAM. By combining experimental data with AI-based prediction engines, the system allows predictive reliability assessment before fabrication, supporting reliability-aware DTCO and industrial deployment in display, automotive, and defense electronics.			
Rationale for Joint R&D Project	Semiconductor devices are rapidly scaling beyond traditional CMOS, whcih urgently need new reliability investigation methods, particularly under: high-energy radiation, mechanical stress, thermal cycling, electromigration, interface degradation. However, in order to build a such reliability investigation platform it requires many innovations, e..g, neutron-based defect visualization, materials and device physics modeling, high-throughput reliability analytics, AI-based prediction of degradation, system-level digital twins, etc. Therefore, they are expensive and very specialized. This project actually directly addresses this critical challenge facing semiconductor, display, automotive, and defense industries: the absence of systematic, physics-driven reliability prediction tools and accessible neutron-based defect analysis infrastructure. Neutron-based reliability evaluation and			

	digital twin modeling require highly specialized infrastructure, multidisciplinary expertise, and international-standard validation, which are currently unavailable in Korea. Collaboration with German research institutes providing advanced NDT methodologies, multiscale reliability testing under external stress, experimental database for material reliability models and digital twin validation frameworks that cannot be independently developed in a short time frame. This Korea–Germany joint R&D is essential to close the national reliability infrastructure gap, accelerate technology development, and establish a globally competitive reliability platform aligned with international standards.
Key Words	1. Digital Twin Reliability 2. Neutron-Based Non-Destructive Testing 3. AI-Driven Reliability Prediction 4. Semiconductor Device Degradation 5. Reliability-Aware DTCO

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The project aims to develop a reliability platform integrating a compact neutron-beam non-destructive testing (NDT) system, and AI-driven reliability analytics and digital twin based on real experimental measurements for emerging semicondutor devices.			
R&D Target Specifications	-The developed system will enable precise, laboratory-scale neutron irradiation for radiation reliability evaluation in semiconductor, AI, and hardware security applications. High-density plasma ion source for stable deuterium ion generation, high-voltage insulated acceleration structure with integrated thermal management and cooling design and proven high-temperature and high-voltage compatible feedthrough technologies will be ensured. A in.house design and fabrication of high-voltage power supplies exceeding 100 kV will be achieved. A maximum neutron output of up to 1×10^8 n/s will be achieved. -In the materials part, several targets should be achieved: (1) wafer-scale 2D epitaxial growth (2–6 inch) with uniform TMDC films (MoS_2, WS_2, $MoTe_2$), (2) Atomic-layer thickness control (±1 ML) enabling bandgap and carrier-type engineering, (3) In-situ doping and alloying ($Mo_{1-x}W_xS_2$) tuning Fermi-level and threshold-voltage, (4) Hetero-epitaxial stacking (2D/oxide, 2D/2D) controling van der Waals superlattice structures, (5) Surface-roughness control (RMS <0.3 nm) enabling ultra-low contact resistance interfaces. -1/f noise extraction and trap model validation for Oxide TFTs and 1D, 2D semiconductor based TFTs. -Device reliability measurements will include photo-assisted BTS/LBTS analysis, high-voltage stress (±100 V) suitable for oxide TFTs and power devices, trap-sensitive 1/f noise extraction by ultra-low current resolution (1 pA) and early-stage reliability degradation detectionable long-term operation. -Fabrication and optimization of hafnia-based ferroelectric devices with emphasis on switching speed, endurance, retention, and variability. -Expansion of ferroelectric and charge-trap devices into vertically stacked crossbar and transistor-based arrays, targeting ≥24×24 array size with more than four stacked layers. Comprehensive array-level measurement and benchmarking. -Device and array reliability analysis will incorporate wafer-scale 2D material integration and advanced array device engineering. -Validation of CIM/PIM and neuromorphic AI applications, including PUF, TCAM, and SNN. -Advanced multiscale reliability testing including in-situ measurements under thermal, mechancial and electrical stress. NDT using X-ray and high resolution analysis (down to atomic level) in microscopes will cover the scale from package level, device level down nanoscale to investigate the defects/failures. Materials' property database will be obtained for creating proper material reliability models. -AI-driven material model and digital twin development.			
R&D Plan	WP1: Develop and build the compact neutron generation device-based NDT system. It includes neutron source engineering, system design, hardware installation and integration, system miniaturization, interface development, etc. (Month 1 to 30) WP2: Wafer-level 2D material growth with process optimization. (Month 1 to 18) WP3: Building 2D materials' reliability database. In particular, the reliability data/quality under stress and radiation conditions will be generated, which will be used to train the AI-based reliability prediction models. (Month 13 to 32) WP4: Emerging semiconductor device fabrication based on wafer-scale 2D materials, e.g, memory and neuromorphic devices, etc. (Month 4 to 12) WP5: Evaluation of device and related architectures towards neuromorphic applications,			

	establishment of reliability metrics under stress and neutron radiation for predictive reliability model. (Month 7 to 30) WP6: Device reliability measurement focusing on electrical noise analysis under low-temperature and light-induced degradation, and physics-based modeling, leading to early failure signature extraction. Development of AI-ready datasets and predictive reliability models that link electrical behavior to neutron- and radiation-induced defect evolution. (Month 4 to 33) WP7: Multiscale (nanoscale to device-level) reliability testing under external stress, creation of experimental database for material reliability models. Correlation with elemental and microstructural information obtained by X-Ray NDT and high-resolution microscopic analysis. Quantitative analysis of neutron-beam radiation generated defects. (Month 1 to month 30) WP8: Material model and digital twin development, validation of digital twin predictions. Integration of neutron-based non-destructive testing system into AI-driven digital twin reliability platform. (Month 13 to 36) WP9: Project management including aligning with EU industrial guidelines. (Month 1 to 36)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The participating research institute will serve as the scientific backbone and validation authority in this project. (i) Multiscale (nanoscale to device-level) reliability testing under external stress, correlated with microstructural changes. Experimental database for material reliability models. (ii) X-Ray/NDT and high-resolution microscopic elemental and microstructural analysis of material/device failure/defects, in particular neutron-beam radiation generated nanoscale defect evolution. (iii) Support material model and digital twin development. Validation of digital twin predictions. (iv) Align methodologies with EU industrial standards, and ensure international credibility of the integrated reliability platform.
Expected R&R of Korean SME Partner	The Korean SME partners will act as technology implementers and industrialization drivers, translating research outcomes into deployable platforms, and bridge the gap between academic reliability science and real-world semiconductor, display, automotive, and defense applications. (i) Leading the development of a compact, industry-deployable neutron-beam NDT system. This includes neutron source engineering, system miniaturization, safety compliance, and interface development for integration with AI-driven reliability analytics and digital-twin frameworks. (ii) Wafer-scale 2D material growth, process optimization, and reliability database construction. Extending from device-level characterization to array-level integration. Building statistically meaningful epitaxy data under various stress and radiation conditions, enabling realistic training and validation of AI-based reliability prediction models. (iii) Device-level reliability characterization and physics-based modeling. Developing AI-ready datasets and predictive reliability models. (iv) Evaluation of emerging memory and neuromorphic devices and related architectures. Establishing reliability metrics under radiation and stress conditions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Outcomes** (1) World's First AI-Driven Neutron-Aware Digital Twin Reliability Platform (2) Wafer-Scale Reliability Database for Emerging Devices (3) Compact Neutron-Beam NDT System for Industrial Deployment (4) Physics-Driven Reliability Prediction Engine (5) Internationally Validated Reliability Standards **Industrial and Societal Impact** **-Strategic Leap in Korea's Semiconductor Reliability Infrastructure** Close a decade-long infrastructure gap by establishing Korea's first neutron-based reliability evaluation and prediction capability. Position Korea as a global leader in reliability-aware semiconductor design (DTCO). **-Acceleration of Time-to-Market and Cost Reduction** Enable predict-before-fabrication workflows, reducing trial-and-error prototyping, lowering development costs, and shortening product cycles for SMEs and large enterprises alike. **-High-Reliability Electronics for Automotive, Defense, and Space** Provide mission-critical reliability assurance under critical conditions required for automotive electronics, aerospace systems, and defense applications. **-Strengthening Korea–Germany Technology Sovereignty** Establish a long-term Korea-Germany reliability alliance, combining Korea's device and manufacturing strengths with Germany's leadership in NDT and reliability engineering. **-SME-Driven Commercialization and Global Market Entry** Empower Korean SMEs to commercialize neutron-beam NDT systems, reliability software, and digital-twin platforms, creating export-ready products and high-value jobs. **-Foundation for Next-Generation AI and Neuromorphic Systems** Enable reliable deployment of AI hardware and neuromorphic devices by ensuring lifetime predictability and radiation robustness at the materials, device, and system levels.

6. References

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization